Real or not?

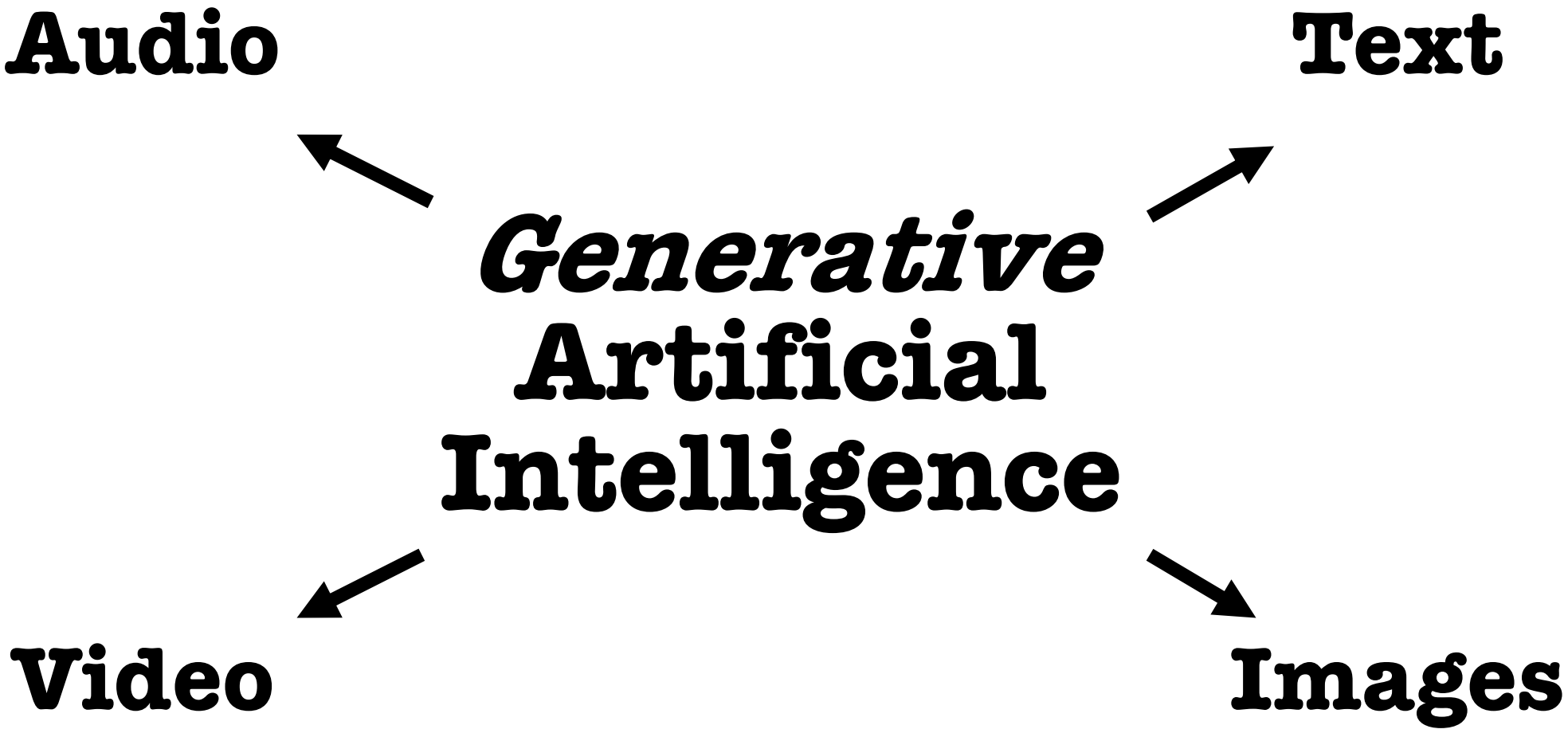








Generative Artificial Intelligence


Audio
Text
Generative
Artificial
Intelligence
Video
Images

Text

Text: Embeddings

King = Queen – Woman + Man

	Animal	Fluffy	Pet	Dangerous	Food
Cat	.98	.94	.87	.30	.15
Dog	.98	.87	.95	.35	.15
Pig	.95	.20	.45	.26	.60
Carrot	.01	.02	.01	.01	.97
Rock	.01	.02	.43	.25	.01

How a language model writes: one word at a time

Every word an AI produces is chosen by repeating these three steps.

1. The text so far

for | the | glory | of | Old

already turned into numbers

2. It looks back at what came before



thicker arrow = pays more attention
repeated across dozens of layers

3. It scores every word

State	48%
Dominion	9%
Home	6%
Timer	5%
Age	4%

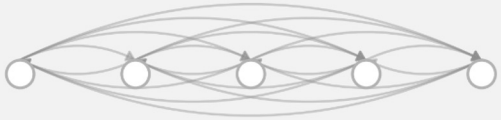
...plus 50,000 more

Pick one · add it to the sentence · start over

"for the glory of Old State" — and now the model asks itself the very same question again.

ENCODER

reads the whole passage at once



Every word can see every other word, in both directions. Built for understanding text — translation, search, sorting, sentiment.

DECODER

writes the next word, over and over



Each word can see only the words before it; it cannot peek ahead. ChatGPT, Claude and Gemini are all built this way.



Click to see full view

Ask Alexa

- Can it be used in saltwater?
- What material is the housing?
- Sponsored Why choose Dr.Fish barrel snap swivels?
- Is this swivel good for bass fishing?
- Why you might like this
- Ask something else

Dr.Fish 30 Pack Barrel Snap Swivels, Barrel Swivels Fishing Tackle Crane Rolling Coast Lock Snap Lure Leader Line Connector Fresh&Saltwater

Visit the Dr.Fish Store

4.6 ★★★★☆ (230) | Search this page

$5.99 Price history

Unlock a $150 Amazon Gift Card upon approval for Prime Visa.

prime

FREE Returns

Size: **#12-13LB-30 Pack**

#12-13LB-30 Pack	**#14-9LB-30 Pack**
$5.99 FREE Delivery **Friday**	$5.49 FREE Delivery **Saturday**

$5.99 Price history

prime

FREE delivery **Friday, September 4**

Deliver to Christo... - State C... 16803

Quantity: 1

Add to cart

Buy Now

Ships from	Amazon
Sold by	Dr.fish
Returns	FREE 30-day refund/replacement
Gift options	Available at checkout

See more

Add to Auto Buy

Add to List



Jason Jones

★★★★★ **Great swivels for easy lure swaps**

Reviewed in the United States on May 17, 2026

Size: #10-18LB-30 Pack | **Verified Purchase**

My go to swivel. I like to change lures and colors A LOT, so yeah, I run a swivel. These are small enough to not effect the action of the lure, yet stronger than the line I tie them to. Work great if you like to change things up a bunch.

Helpful | Report

AI

Mark Sullivan

★★★★★ **Well-made swivels and a good value**

Reviewed in the United States on June 8, 2026

Size: #10-18LB-30 Pack | **Verified Purchase**

The swivels appear to be well made and work exactly as expected, with a smooth rotation and a secure snap that makes changing lures or leaders quick and easy. The 30-pack is a good value, and the assortment of sizes makes these useful for a variety of fishing applications. They seem sturdy enough for regular use and are a handy addition to any tackle box.

Helpful | Report



Alex Sush

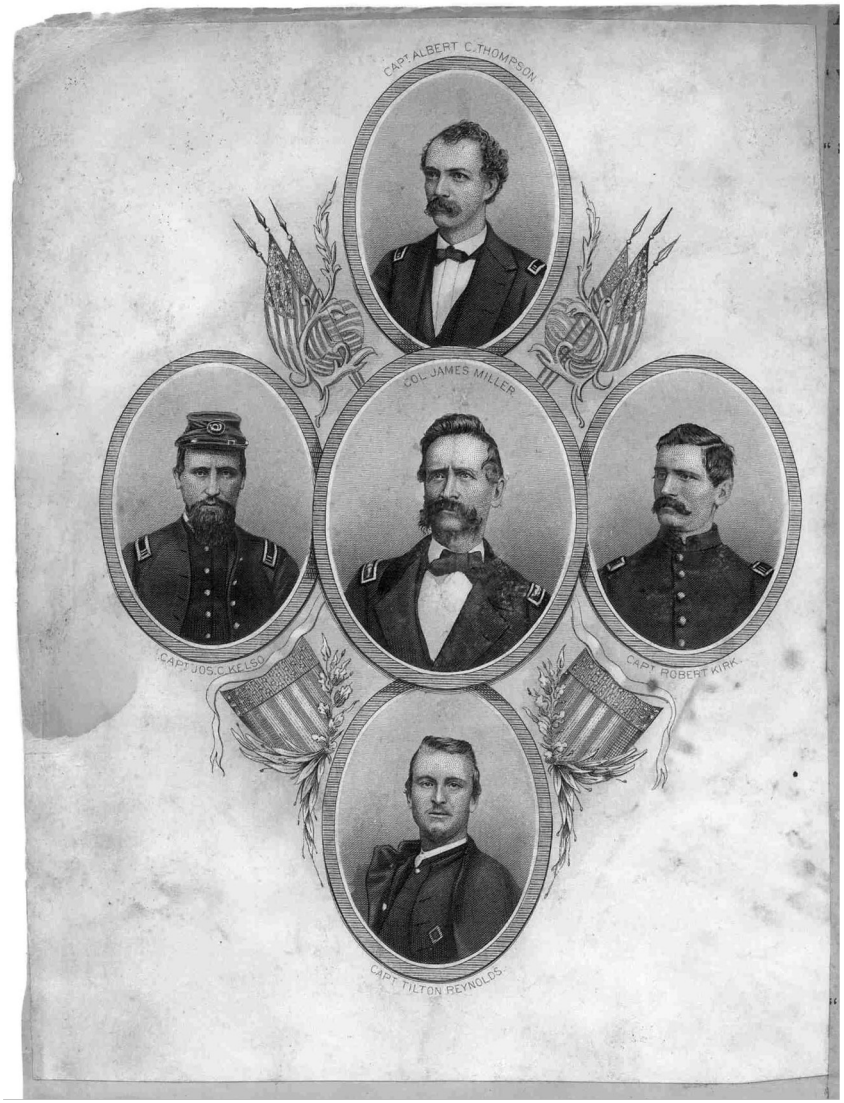
★★★★☆ **Good but should cost less**

Reviewed in the United States on November 24, 2020

Verified Purchase

Quality is good. The angle that is snaps at is weird, sometimes hard to use it. Swivel itself decent quality, spins freely. The sizing of the whole snap swivel is slightly off comparing to other big brands, it runs a bit small. Pulled out 7 lbs carp with no issues. Overall it's a good buy, although it should cost $5 not $8.

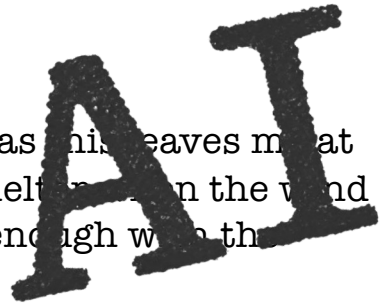
Helpful | Report


CAPT. ALBERT C. THOMPSON.
COL. JAMES MILLER
CAPT. JOS. C. KELSO
CAPT. ROBERT KIRK.
CAPT. TILTON REYNOLDS.



Camp Griffin Jan 9th 1862

Dear Mother,



I received your letter of the 2nd and was glad to hear you are all in good health as this leaves me at present. We have had a hard week of it here with the cold. The tents are poor Shelter when the wind gets up in the night and a good many of the Boys have taken cold. I keep warm enough with the Blankets you Sent and the overcoat So you need not worry on that acct.

There is not much doing here. We drill in the fore noon when the ground will bear It and the rest of the time a man has to find his own amusement. Some of the Boys have got up a Debating Society and meet in the Chaplains tent of an evening. I went once But it was mostly the Same three or four talking and I had rather write letters.

Our rations are Better than they were. We drew fresh Beef twice this week which is a great improvement on the Salt. I have not been Sick a day Since I came out which is more than most can Say.

Jacob Wolfe got a Box from home and there was a fruit cake in It that had gone Bad. We ate the rest of It any way. If you Send one do not put in any thing that will not keep for they are Slow in coming.

Write when you can and tell me how Father is getting on with the Stock.

Your affectionate Son
Tilton

Headquarters 105th Reg't. Co. H
Camp Jameson Nov 24th 1861

Dear Mother,
Again I take up my pen to write a few lines to you though I have told Clarinda nearly all t[illegible] news I could [illegible]k off. But if have no news I will write any way. Bill Green has Spent nearly all the money he ha[illegible]eft S[illegible]r[illegible]a poor Sight of him Sending any to you which he Should have done in the first place...There is on[illegible]hing I[illegible] been goin to tell you for a long time that is concerning our Col. I never Seen a man So Universaly hate[illegible]s he is. There is no[illegible] man in the Regt that likes him. He is hard on his men puts them on the Ring (ie) (a circle for[illegible]m to walk rou[illegible]in with a guard in the middle with his Bayonet to make them travel) for every little thing they do. [illegible]d to circulate this or let any one know from what Source you got it But it is a fact he Stands too much on his Dignity. Some of the Capts here (if they could) would take their companies away in a Short time. But this will do for the Col Business.

You Said if you could you would keep that money until I came home. Now you need not do any Such a thing. I Sent It Because I wanted you to have It to use for Such things as you need. So you will not Save It any on my acct. Uncle Orlando is writing a letter to Aunt Rebecca. We are going to get a Store tomorrow and then we can live as comfortable as you please. I thought Some of getting my Likeness taken large on paste Board and put in a frame and Send It home By Express. How would you like It? That way you could hang It up in the room or some place where you could see it any time. I have Seen Several taken on cards they are verry nice too. I will in my next letter Send you a likeness of Gen Jameson (our Brig Gen). It will be out next week. He is a fine looking man. I will also send you a likeness of Gen Heintzelman (Maj Gen) who commands our division.

But I must quit for it is getting late. Good By Dear Mother from your affectionate son.
Tilton

Headquarters 105th Reg't. Co. H
Camp Jameson Dec 1st 1861



Dear Mother,

I got your letter of the 26th and was glad to hear you are all well as this leaves me at present. You wanted to know about my Boots. The pair I brought from home are nearly done for the Soles have parted from the uppers on the right one and there is a poor Sight of them lasting out the month. There is no mending them here that is worth any thing. The Sutler has a fair article at $4.00 which is a verry great price But the Shoes the government gives out are miserable Stuff and a man might as well go Bare foot in this mud. Uncle Orlando Says I had Better pay the money once and be done with It than to Buy twice.

I have $6.50 By me yet after paying Wm Hays what I owed him and Some little dribs Besides So I can manage It without asking you for any thing. Now I dont want you to Say any thing to Father about what I gave for them for he will think I have been foolish and there is no use in It.
Some of the Boys Say they will go to Alexandria for their Boots But I think By the time they pay for the going they will Save nothing. But this will do for the Boot Business.

Uncle Orlando is well and Sends his respects. I have not heard from Clarinda this week.
But I must quit for it is getting late. Good By Dear Mother from your affectionate son.

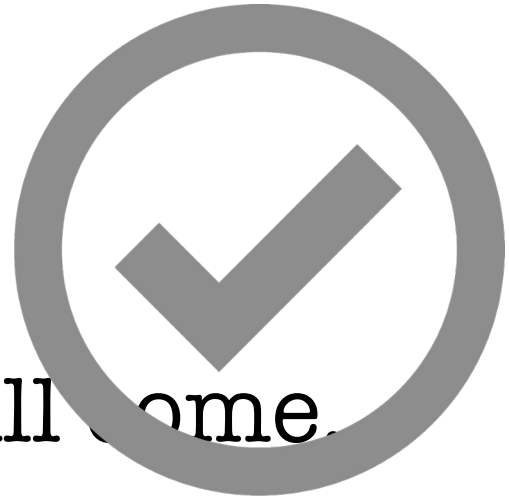
Tilton

There is a quiet hush that falls,
When evening shades draw near,
A stillness in the air that calls,
To those who pause to hear.

The world around us fades away,
As darkness takes its hold,
And in this moment, we may pray,
Or simply sit and fold.

For in this hush, we may find peace,
And let our worries go,
And in the silence, we release,
All that we cannot know.





Not knowing when the Dawn will come.
I open every Door,
Or has it Feathers, like a Bird,
Or Billows, like a Shore —

I speak of love that comes to mind:
The moon is faithful, although blind;
She moves in thought she cannot speak.
Perfect care has made her bleak.

I never dreamed the sea so deep,
The earth so dark; so long my sleep,
I have become another child.
I wake to see the world go wild.

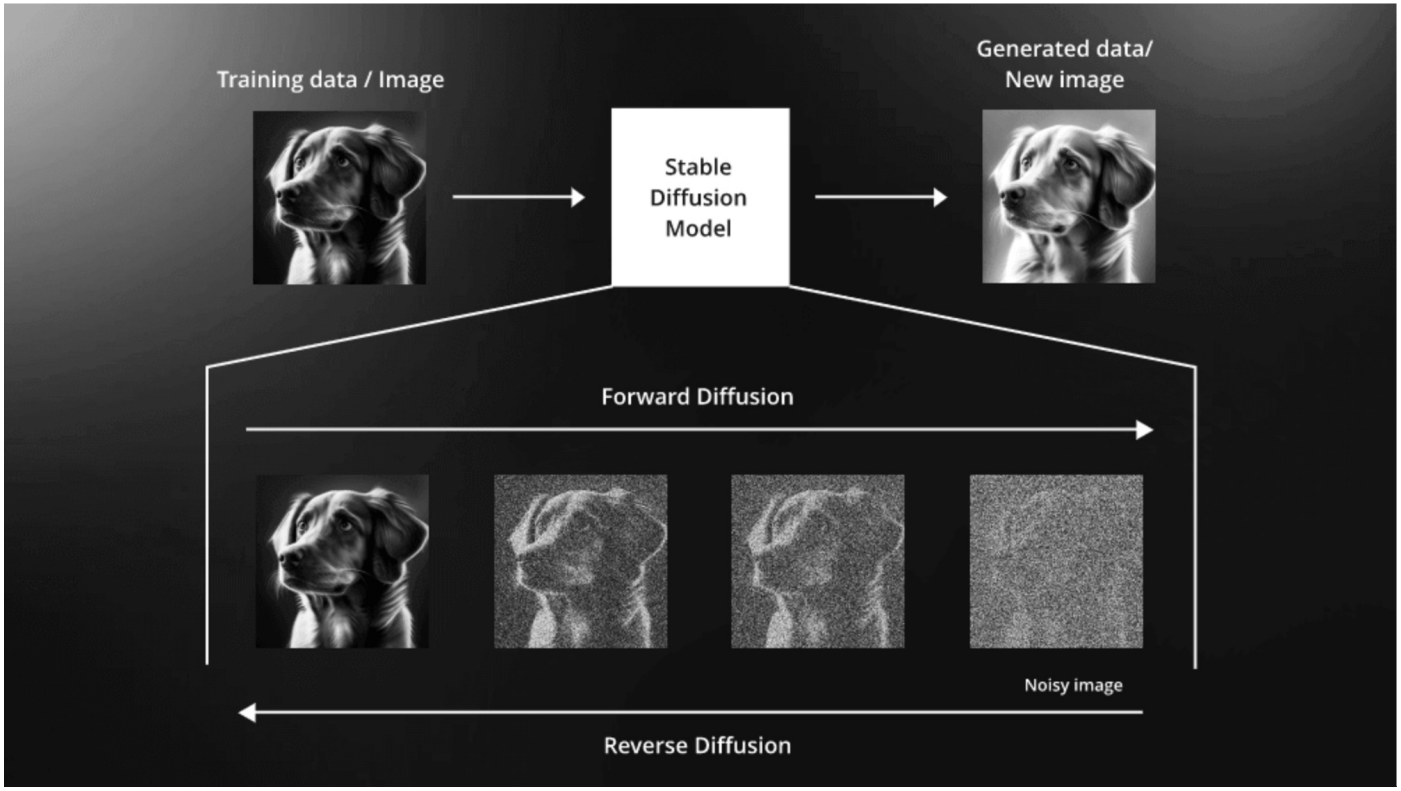




Pigeons shake their wings on the copper church roof
out my window across the street, a bird perched on the cross
surveys the city's blue-grey clouds. Larry Rivers
'll come at 10 AM and take my picture. I'm taking
your picture, pigeons. I'm writing you down, Dawn.
I'm immortalizing your exhaust, Avenue A bus.
O Thought, now you'll have to think the same thing forever!

Images

Images: Diffusion Models



https://poloclub.github.io/diffusion-explainer/









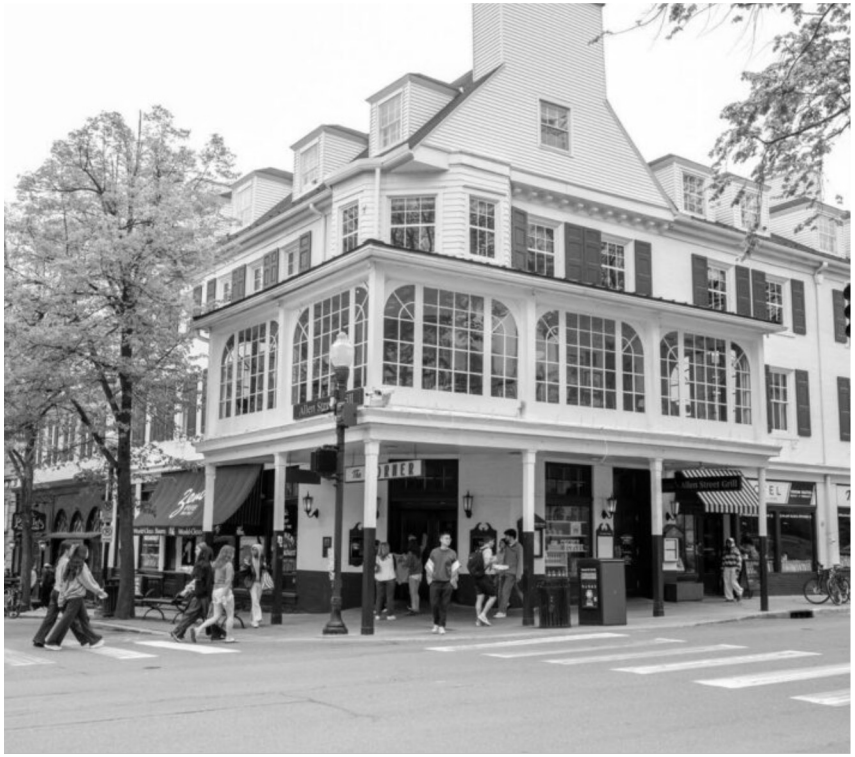




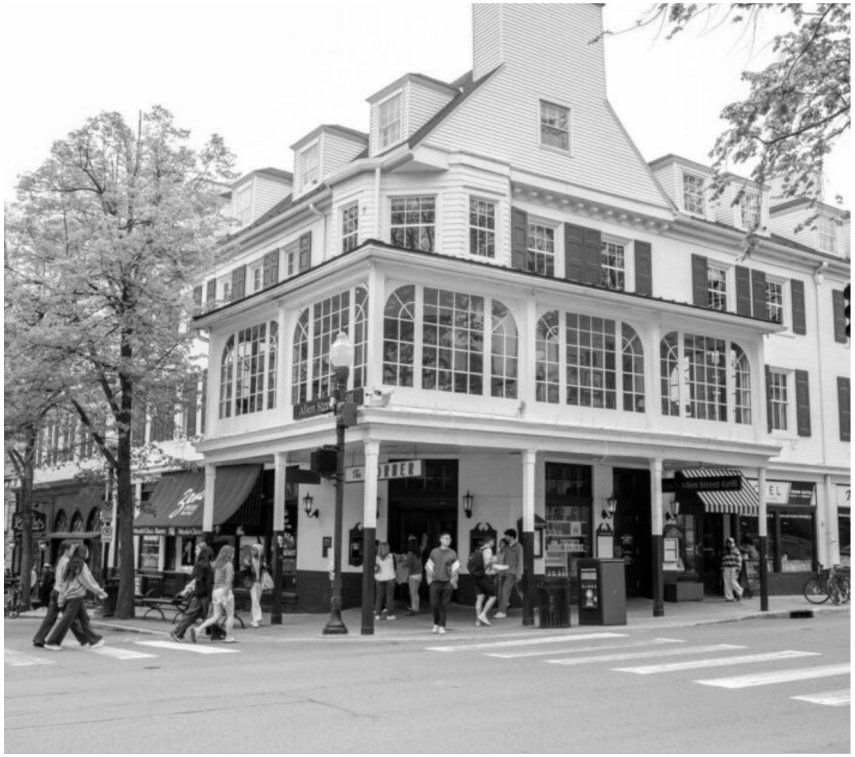




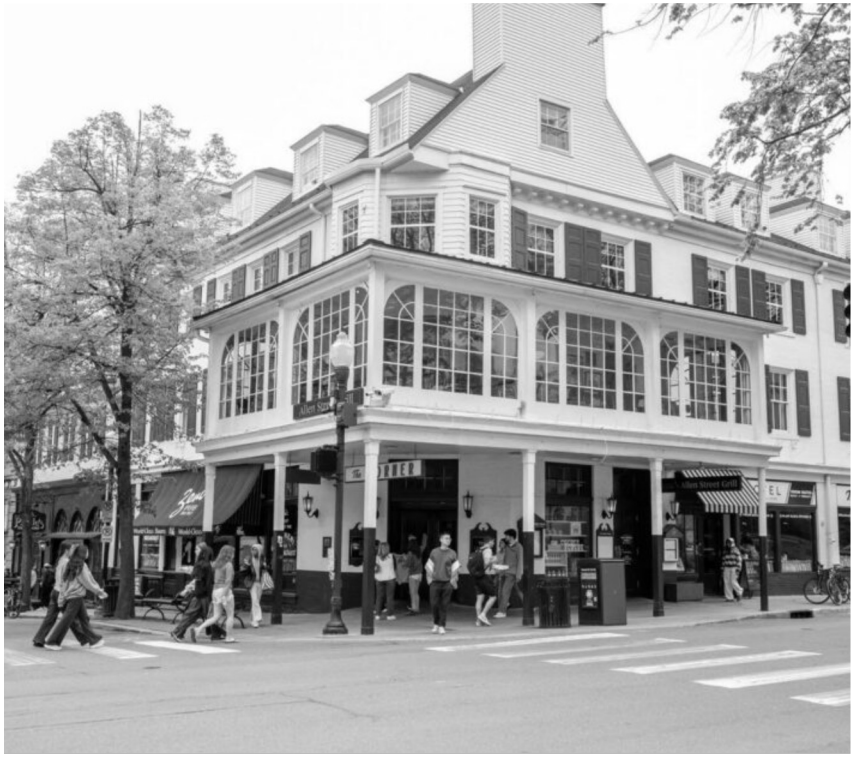

CORNER ROOM
- STATE UF.L E, PA
CORNER ROOM
- JOALE, PA,
STATE COLLEGE
yeri.ai




THE CORNER ROOM
THE CORNER ROOM
Allen Street Grill



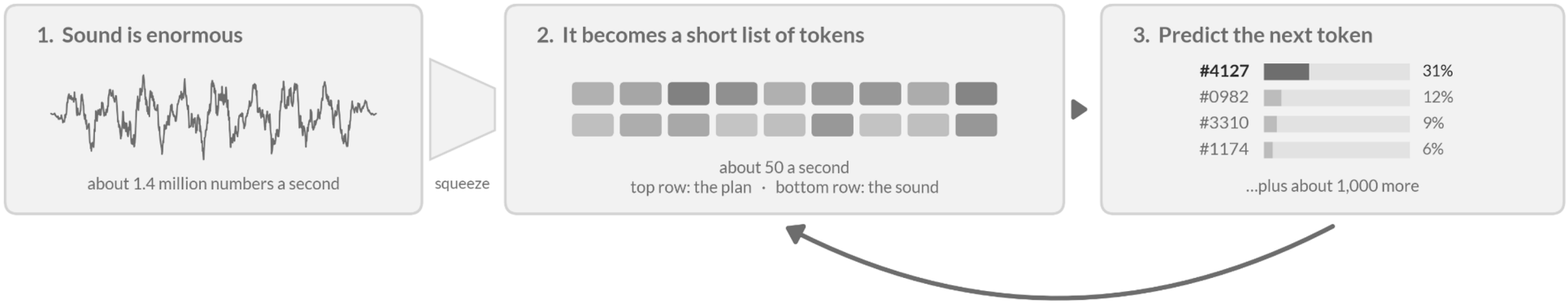

THE LION'S SPOT
THE LION'S SPOT



Audio

How AI makes music: the same loop, a different alphabet

Sound is far too big to predict one number at a time — so it is squeezed into tokens first.



Pick one · add it to the sequence · start over

The identical loop from the text model — only the alphabet has changed.

THE PLAN

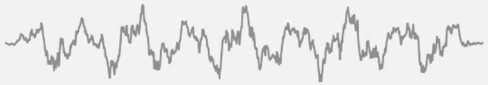
semantic tokens



Where the chorus falls, how the melody moves, what words get sung. The structure that holds a song together.

THE SOUND

acoustic tokens



The timbre of a voice, the room, the hiss of the recording. Detail with no plan behind it wanders off within seconds.

BACK TO AUDIO

the decoder



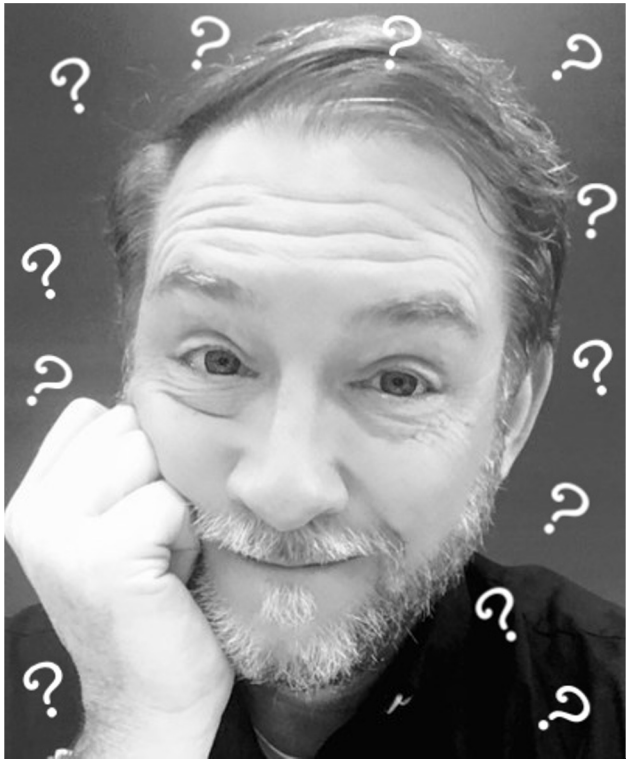
The squeeze runs in reverse, expanding the tokens into a file you can play — 1.4 million numbers a second once more.

Architecture as published for research systems (AudioLM, SoundStream/EnCodec) and for Suno's open-source Bark. Commercial music services do not disclose their designs.









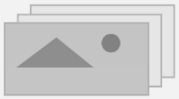
https://sightengine.com/ai-or-not-music

Video

How AI makes video: the whole clip at once

Not a filmstrip built frame by frame — a solid block of frames, carved out of noise together.

1. A clip is enormous



ten seconds of HD is about a billion numbers

2. Squeezed in three directions



narrower, shorter — and fewer frames

3. The block starts as pure noise



every frame at once — none of them first

4. Then the entire block is cleared — together

no first frame · no next frame



about forty passes — each one improves every frame simultaneously

SPACE

attention inside each frame



Does this single picture hold together — the light, the shadows, the reflections? This is where a still image goes wrong.

TIME

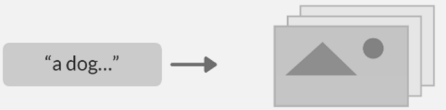
attention across frames



Does the motion hold together — does the cup stay on the table? Handled by its own separate layers, not as an afterthought.

THE PROMPT

words steer every pass



Your words run through a language model first, then guide all forty passes. The same machinery you saw on the text slide.

Architecture as published for Seedance 1.0 (ByteDance) and open research systems. Veo, Sora and Kling do not disclose their designs. Step counts are illustrative.


Will Smith Eating
Spaghetti Progression
2023
2024
2025
2026



AI





AI


GPO
we have achieved tremendous



AI

So?...

- Discernment – pay attention
- How should we know what we know?
- Avoid being discouraged...

Thank you!

Christopher Zorn
zorn@psu.edu

My sources:

- "AI or Not?" (faces)
- Not Knowing When The Dawn Will Come
- An Eastern Ballad
- "AI-Generated Poetry is Indistinguishable from Human-Written Poetry and is Rated More Favorably," *Nature Scientific Reports* (2024)
- Dr. Fish Snap Swivels (on Amazon)
- Civil War Soldier in the Wild Cat Regiment: Selections from the Tilton C. Reynolds Papers (Library of Congress)
- Diffusion Explainer (MIT Polo Club)
- "AI or Not? (music)
- The Will Smith Eating Spaghetti Test (Wikipedia)
- Bunnies Jumping on a Trampoline (TikTok)
- Nittany Lion @ the Pool (YouTube)
- Netanyahu Remarks (YouTube)
- ICE video (Instagram)

My tools:

- Claude Opus 5 (general use + text)
- Midjourney (images)
- Suno 5.5 (music)
- Seedance 2.5 (videos)
- Downr.org (video downloads)